Nikhilesh Vijay Chirutala

Full-Stack Software Engineer

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SUMMARY

Software Engineer with 3 years of professional experience building and maintaining production applications and backend services across cloud contact-center, CRM, fintech, and AI platforms. Experienced in Python, FastAPI, REST APIs, microservices, AWS, data processing, cloud-storage workflows, and production operations. Migrated a legacy call-recording workflow to a Python service deployed across 40 production clusters and built validation, retry, recovery, and idempotent storage workflows. Focused on reliable service behavior, debugging, deployment validation, and observability using Datadog, Grafana, and Sentry, with additional end-to-end application development experience.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Backend & Frameworks: Python Services, FastAPI, Flask, REST APIs, API Development, Backend Automation, Microservices

Frontend: React.js, Next.js, Tailwind CSS, Bootstrap, Jinja2

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

Databases: MongoDB, PostgreSQL, Neon DB, SQL

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux, Production Deployments, Monitoring

APIs & Integrations: Twilio, GitHub API, REST APIs, Postman, API Integration

Security & Authentication: Authentication, Authorization, RBAC, ACL, JWT

Testing & Tools: Postman, Jest, Unit Testing, Git, GitHub, Debugging, Deployment Validation, Root-Cause Analysis

WORK EXPERIENCE

Software Engineer · **CA One Tech Cloud Inc.** Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform *Python, FastAPI, AWS, Amazon S3, MySQL, Datadog*

- Develop and maintain production-grade Python services for a high-volume cloud contact-center platform, supporting call-recording ETL, data processing, storage workflows, and production operations with a focus on reliable execution.
- Migrated a legacy Perl-based call-recording workflow to a long-running Python service deployed across 40 production clusters, using feature flags, staged rollout, validation, rollback, and monitoring to manage production changes.
- Built Python integrations with a Call Intelligence microservice, implementing service APIs, eligibility checks, account and campaign controls, daily processing limits, retries, response validation, and result persistence for dependable service-to-service processing.
- Developed validation and recovery workflows for call-recording data, handling incomplete processing, failed operations, and inconsistent storage states across production environments to improve operational reliability.
- Supported production deployment and rollout of Python services across multiple clusters, including feature-flag configuration, validation, monitoring, and troubleshooting during migration activities.
- Built AWS S3 audit and recovery workflows for recording storage, including deterministic object handling and idempotent re-upload operations for incomplete or failed storage migrations, supporting safe recovery and repeatable processing.
- Implemented production observability using Datadog, Grafana, dashboards, and Sentry to monitor service health, track errors, diagnose processing failures, and support production incident resolution.
- Investigated production issues across Python services, APIs, databases, and cloud storage, performing troubleshooting and implementing fixes to maintain reliable recording-processing workflows.

Software Engineer · **Mantra Technologies** Oct 2023 – Mar 2026 | Hyderabad, India

- Delivered production features and enhancements across multiple applications, translating product requirements into application modules, business logic, and complete end-to-end workflows.
- Worked across application layers to implement and integrate functionality, connecting frontend workflows, backend services, APIs, and data flows to deliver cohesive product features.
- Implemented authentication, authorization, and access-control workflows, while integrating external services and handling validation and integration-related issues.
- Supported production releases and resolved application issues through testing, debugging, root-cause analysis, deployment validation, and defect resolution, contributing to the software delivery lifecycle.

Key Projects:

AI Bull — AI-Powered Trading & Investment Platform *React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM*

- Developed an AI-powered trading and investment platform with portfolio aggregation, enabling users to consolidate holdings from multiple Demat accounts such as Upstox, Zerodha, and Groww into a unified view.
- Built portfolio workflows for tracking positions, stocks, and mutual funds, along with detailed reports and investment-level information across aggregated accounts.

- Developed stock and mutual fund screeners with configurable filtering capabilities and detailed asset pages for analyzing individual investments and their underlying information.
- Designed and implemented a backtesting engine for evaluating trading strategies against historical market data and analyzing strategy performance.
- Implemented a Retrieval-Augmented Generation (RAG) based chatbot that combines portfolio and financial information to provide contextual insights and answer portfolio-related user queries.
- Built FastAPI and Python backend APIs and business logic supporting portfolio aggregation, investment analysis, screening, backtesting, and AI-powered portfolio workflows.

CRM.io — AI-Driven CRM System *FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, WebSockets, AWS S3, AWS Lambda, FCM*

- Developed backend modules and REST APIs for an AI-driven CRM using Python and FastAPI, implementing business logic and workflows across Contacts, Tasks, Deals, Documents, communication, and account management.
- Built the Contacts management experience with inline editing, global search across CRM modules, access to related Tasks, Deals, and Documents, and Google Sheets integration for importing contact records in bulk.
- Developed a real-time CRM dialer accessible from contact records, integrating Twilio and an internal SIP-based telephony application through WebSockets, with AI-generated call scripts provided for relevant calls.
- Implemented a CRM-wide soft-delete and Trash workflow with a 30-day retention period, replacing hard-delete behavior and using an AWS Lambda service to permanently remove records after the retention period.
- Built document management workflows using Amazon S3, including browser-to-S3 client uploads and storage of documents associated with CRM records.
- Developed a task cadence system that allows users to create sequences of tasks as a single workflow and progress through scheduled follow-ups, with Firebase Cloud Messaging (FCM) used for task notifications.
- Built account-level reporting dashboards and developed an MCP integration exposing CRM capabilities as tools, enabling the AI bot to retrieve CRM information and trigger supported operations.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground *Next.js, GitHub API, Vercel*

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform *Next.js, Neon DB, Clerk, Vercel*

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

EDUCATION

B.Tech — Electrical & Electronics Engineering 2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur